

# Backend Schema Document

**Version:** 0.1 (draft). Postgres + PostGIS (Supabase). Adjust names and types to taste.

## 1. Entity Overview



## 2. Enums and Extensions

```
sql

create extension if not exists postgis;
create extension if not exists pgcrypto;

create type user_role as enum ('hotel','ngo','volunteer','public','admin')
create type org_status as enum ('pending','approved','rejected','suspended')
create type venue_type as enum ('hotel','restaurant','banquet_hall','hostel')
create type food_diet as enum ('veg','non_veg','mixed');
create type listing_status as enum ('posted','claimed','completed','expired','c
create type pickup_status as enum ('claimed','on_the_way','collected','complet
create type tag_kind as enum ('positive','issue');
create type flag_status as enum ('open','kept','corrected','voided');
create type news_status as enum ('draft','published','hidden');
create type news_source as enum ('internal','external');
```

## 3. Core Tables

```
sql
```

```

create table profiles (
  id          uuid primary key references auth.users(id) on delete cascade,
  role        user_role not null,
  full_name   text not null,
  phone       text,
  created_at  timestamptz not null default now()
);

create table hotels (
  id          uuid primary key default gen_random_uuid(),
  owner_id    uuid not null references profiles(id),
  name        text not null,
  slug        text unique not null,
  venue_type  venue_type not null,
  address     text not null,
  location    geography(Point,4326) not null,
  rooms       int default 0,
  seats       int default 0,
  banquet_capacity int default 0,
  events_per_month int default 0,
  operating_days_per_week int default 7 check (operating_days_per_week between
status      org_status not null default 'pending',
public_listing boolean not null default true,
verification_doc_path text,
created_at   timestamptz not null default now()
);

create table ngos (
  id          uuid primary key default gen_random_uuid(),
  owner_id    uuid not null references profiles(id),
  name        text not null,
  registration_no text,
  address     text not null,
  location    geography(Point,4326) not null,
  capacity_portions int default 0,
  service_radius_km numeric default 10,
  trust_score  numeric not null default 1.0, -- 0..1, lowered by ou
status      org_status not null default 'pending',
verification_doc_path text,
created_at   timestamptz not null default now()
);

create table volunteers (
  id          uuid primary key default gen_random_uuid(),

```

```

profile_id  uuid not null references profiles(id),
ngo_id      uuid not null references ngos(id),
vehicle     text,
active      boolean not null default true
);

create table listings (
  id          uuid primary key default gen_random_uuid(),
  hotel_id    uuid not null references hotels(id),
  title       text not null,
  description  text,
  diet        food_diet not null,
  portions_listed int not null check (portions_listed > 0),
  cooked_at   timestampz not null,
  pickup_by   timestampz not null,
  packaging_notes text,
  safety_confirmed boolean not null default false,
  status       listing_status not null default 'posted',
  created_at   timestampz not null default now(),
  check (pickup_by > cooked_at)
);

create index on listings (status, pickup_by);
create index on listings (hotel_id, created_at desc);

create table pickups (
  id          uuid primary key default gen_random_uuid(),
  listing_id   uuid not null references listings(id),
  ngo_id       uuid not null references ngos(id),
  volunteer_id uuid references volunteers(id),
  status       pickup_status not null default 'claimed',
  claimed_at   timestampz not null default now(),
  started_at   timestampz,
  collected_at timestampz,
  completed_at timestampz,
  collected_in_geofence boolean,
  portions_received int check (portions_received >= 0),
  left_behind_portions int check (left_behind_portions >= 0),
  photo_path   text,
  cancel_reason text,
  voided       boolean not null default false,
  created_at   timestampz not null default now()
);

-- Only one active pickup per listing at a time
create unique index one_active_pickup_per_listing
  on pickups (listing_id) where status in ('claimed', 'on_the_way', 'collected');
```

```
create index on pickups (ngo_id, created_at desc);

create table location_pings (
  id          bigserial primary key,
  pickup_id   uuid not null references pickups(id) on delete cascade,
  location     geography(Point,4326) not null,
  accuracy_m  numeric,
  recorded_at  timestamptz not null default now()
);
create index on location_pings (pickup_id, recorded_at desc);

create table tags (
  code      text primary key,          -- e.g. 'ready_on_time','well_packaged','fr
  label     text not null,
  kind      tag_kind not null,
  active    boolean not null default true
);

create table pickup_tags (
  pickup_id  uuid references pickups(id) on delete cascade,
  tag_code   text references tags(code),
  primary key (pickup_id, tag_code)
);
```

## 4. Moderation, Scoring and Incentives

sql

```

create table flags (
  id          uuid primary key default gen_random_uuid(),
  pickup_id   uuid not null references pickups(id),
  reason      text not null,          -- 'quantity_mismatch','missing_photo','outs
  source      text not null default 'auto',  -- 'auto' | 'admin' | 'email_comp
  status      flag_status not null default 'open',
  created_at  timestampz not null default now(),
  resolved_at timestampz
);

create table admin_actions (
  id          uuid primary key default gen_random_uuid(),
  admin_id    uuid not null references profiles(id),
  action      text not null,          -- 'approve_org','suspend_org','void_report'
  target_type text not null,
  target_id   uuid,
  reason      text,
  before_data jsonb,
  after_data  jsonb,
  created_at  timestampz not null default now()
);

create table settings (
  key        text primary key,        -- e.g. 'score_weights','smoothing_k','min_p
  value      jsonb not null,
  updated_at timestampz not null default now()
);

create table score_snapshots (
  id          uuid primary key default gen_random_uuid(),
  hotel_id    uuid not null references hotels(id),
  period_start date not null,
  period_end  date not null,
  completed_pickups int not null,
  portions_rescued int not null,
  rescue_rate  numeric,
  participation numeric,
  listing_accuracy numeric,
  handoff_quality numeric,
  left_behind  numeric,              -- null if no data
  composite    numeric,
  tier          text,                -- 'seed','sprout','canopy','forest'
  rank_in_category int,
  eligible     boolean not null default false,

```

```
    created_at      timestamptz not null default now(),
    unique (hotel_id, period_end)
);

create table certificates (
  id              uuid primary key default gen_random_uuid(),
  hotel_id        uuid not null references hotels(id),
  month           date not null,
  pdf_path        text not null,
  meals           int not null,
  created_at      timestamptz not null default now(),
  unique (hotel_id, month)
);
```

## 5. Public Engagement and News

sql

```

create table news_items (
  id          uuid primary key default gen_random_uuid(),
  source_type news_source not null,
  title       text not null,
  snippet     text,                -- short only; no full copied articles
  source_name text,
  url         text,
  image_url   text,
  status      news_status not null default 'draft',
  pinned      boolean not null default false,
  published_at timestamptz,
  created_by  uuid references profiles(id),
  created_at  timestamptz not null default now()
);

create table follows (
  user_id  uuid references profiles(id) on delete cascade,
  hotel_id uuid references hotels(id) on delete cascade,
  created_at timestamptz not null default now(),
  primary key (user_id, hotel_id)
);

create table kudos (
  id          uuid primary key default gen_random_uuid(),
  user_id     uuid not null references profiles(id) on delete cascade,
  hotel_id    uuid not null references hotels(id) on delete cascade,
  week_start  date not null,
  created_at  timestamptz not null default now(),
  unique (user_id, hotel_id, week_start)    -- one kudos per hotel per week
);

create table notifications (
  id          uuid primary key default gen_random_uuid(),
  user_id     uuid not null references profiles(id) on delete cascade,
  type        text not null,
  payload     jsonb not null default '{}',
  read_at     timestamptz,
  created_at  timestamptz not null default now()
);
create index on notifications (user_id, created_at desc);

create table push_subscriptions (
  id          uuid primary key default gen_random_uuid(),
  user_id     uuid not null references profiles(id) on delete cascade,

```

```
endpoint    text unique not null,  
keys        jsonb not null,  
created_at  timestampz not null default now()  
);
```

## 6. Key Functions

### Atomic claim

sql

```
create or replace function claim_listing(p_listing uuid, pngo uuid, pvolunteer  
returns uuid language plpgsql security definer as $$  
declare v_pickup uuid;  
begin  
    update listings set status = 'claimed'  
    where id = p_listing and status = 'posted' and pickup_by > now();  
    if not found then  
        raise exception 'ALREADY_CLAIMED_OR_EXPIRED';  
    end if;  
    insert into pickups (listing_id, ngo_id, volunteer_id)  
    values (p_listing, pngo, pvolunteer) returning id into v_pickup;  
    return v_pickup;  
end $$;
```

### Nearby listings for an NGO

sql

```
select l.*, h.name, st_distance(h.location, n.location)/1000 as km  
from listings l  
join hotels h on h.id = l.hotel_id  
join ngos n    on n.id = $1  
where l.status = 'posted' and l.pickup_by > now()  
    and st_dwithin(h.location, n.location, n.service_radius_km * 1000)  
order by l.pickup_by;
```

### Weekly score job (outline)

1. For each approved hotel with `(public_listing)` or not (all get a private score), take pickups with `(status='completed' and voided=false)` in the last 60 days.
2. Compute each signal (definitions in the PRD, Section 8).
3. Apply smoothing, composite weights from `(settings)`, per-NGO cap on tag weight.

4. Set `(eligible = completed_pickups >= min_pickups)`; rank within `(venue_type)` category; assign tier.
5. Insert into `(score_snapshots)`. Voiding or correcting a pickup re-runs this for that hotel.

#### **Auto-flag triggers (on report submit)**

- `(portions_received)` differs from `(portions_listed)` by more than a configured percentage.
- `(photo_path)` missing, or `(collected_in_geofence = false)`.
- Report completed more than 4 hours after collection.
- NGO's recent reports are statistical outliers (low trust score).

## 7. Row-Level Security Summary

| Table                               | Hotel                                    | NGO / Volunteer                                   | Public user                                         | Admin |
|-------------------------------------|------------------------------------------|---------------------------------------------------|-----------------------------------------------------|-------|
| profiles                            | own                                      | own                                               | own                                                 | all   |
| hotels                              | own (full), others<br>(public fields)    | public fields                                     | public fields for<br>opted-in                       | all   |
| ngos                                | none                                     | own                                               | none                                                | all   |
| listings                            | own                                      | open listings in radius;<br>own claimed           | none                                                | all   |
| pickups                             | those on own<br>listings (read-<br>only) | own NGO's                                         | none                                                | all   |
| location_pings                      | read for own<br>active pickups           | write for own active<br>pickup, read own<br>NGO's | none                                                | all   |
| flags / admin_actions<br>/ settings | none                                     | none                                              | none                                                | all   |
| score_snapshots                     | own                                      | none                                              | public view (latest,<br>eligible, opted-in<br>only) | all   |
| news_items                          | published (read)                         | published (read)                                  | published (read)                                    | all   |
| follows / kudos                     | none                                     | none                                              | own                                                 | all   |
| notifications                       | own                                      | own                                               | own                                                 | own   |

- Public leaderboard and hotel public pages are served through **views** exposing only safe columns.
- Photos live in a private bucket; access via signed URLs for the involved hotel, NGO and admin.

## 8. Scheduled Jobs

| Job                                                     | Frequency         |
|---------------------------------------------------------|-------------------|
| Expire unclaimed listings past <code>(pickup_by)</code> | every minute      |
| Re-notify / expand radius for unclaimed listings        | every few minutes |
| Release no-show claims                                  | every few minutes |
| Report-due reminders and late flags                     | every 15 minutes  |
| Score snapshots                                         | weekly            |
| Monthly certificates                                    | monthly           |
| Delete location pings older than 30 days                | daily             |
| RSS news fetch to drafts                                | hourly            |

## 9. Seed Data (for development)

- 5 hotels (mixed venue types), 3 NGOs with volunteers, 1 admin, 3 public users.
- Tags: `ready_on_time`, `well_packaged`, `fresh` (positive); `late`, `quantity_short`, `poor_packaging` (issue).
- Settings defaults for weights, smoothing k, min pickups, geofence radius.
- A script that simulates a volunteer moving along a route to test live tracking.

## 10. Open Schema Questions

- Independent volunteers (no NGO)? Would make `(volunteers.ngo_id)` nullable.
- Quantity unit: portions only, or also kg?
- Multi-branch hotel chains: one account with several locations?
- Keep full history of voided reports (recommended) or delete?